Day 25 — SFTP Connection, Gather with Zip, and Zip Compression

Part A — The SFTP Connection

1. Why Do We Use SFTP?

When we want to **integrate with a file-based server**, we go with **SFTP**. In class SFTP was expanded as "Secure File Transfer Protocol", and the point was: **we don't have any local file servers** in the cloud.

> **Small fix:** SAP's documentation expands SFTP as **SSH File Transfer Protocol** — a file-transfer protocol built on the SSH (Secure Shell) network protocol. (It is *not* the same as FTPS, which is FTP over SSL/TLS.) "Secure File Transfer Protocol" is a common informal name, but use "SSH File Transfer Protocol" in interviews.

2. Why SFTP in CPI but a File Adapter in PI/PO?

If you worked with PI/PO, you might know these file adapters: ① File adapter ② SFTP adapter ③ FTP adapter. **But in CPI we don't have all of these file adapters — the adapter we have in CPI is SFTP.**

> **Question:** why do we have a File adapter in PI/PO, and why not in CPI?
> **Answer:** a **File adapter** is nothing but a **local file server**. **PI/PO is an on-premise system**, so it has **local folders** — we can place files in a local folder and pick files from it. **CPI is on the cloud**, so it has **no folder of its own** to place files in — that's why CPI uses the **SFTP adapter** to reach files elsewhere.

	PI/PO	CPI
Where it runs	On-premise	Cloud
Local folders	Yes → File adapter reads/writes them	No
File integration	File, SFTP, FTP adapters	SFTP adapter

The SFTP Folder Must Be Exposed

Scenario: if we want to place any file in an SFTP folder, **the other side has to expose that folder as an SFTP server folder.** Unless they expose that folder as an SFTP server, **we cannot place or retrieve files** in it. The SFTP server **can be anywhere — on-premise or cloud.**

3. SFTP Adapter — Connection Settings

Field	Example from Class
File Name	`Test.xml`
Folder (Directory)	`/output`
Server host : port	`tcs265.files.com:22` — **port 22** is the standard SSH/SFTP port
Authentication	User name / password (static) · **Public Key** · **Dual Authentication**

4. Public Key–Based Authentication

Certificates → key pair → 2 types of keys: Public key and Private key.

Key	Rule
Public key	This key **will be shared with everyone**
Private key	**Cannot be shared** — it stays with you

The two keys are **interlinked**. **Scenario: we share the public key with the SFTP team**, and **we use our**

private key to send the request. They **verify whether the request is coming from a valid source or not**.

> **Dual Authentication** = using a **combination of both** the public key **and** the username/password.
> **Remember:** as a middleware consultant, **connectivity sits inside the security concern — security is the most important point.**

How to Generate the Key Pair (SSH Key)

1. **Monitor tab → Manage Security → Keystore → Create** → choose **SSH Key** (key pair).
2. Fill in the alias and details, then **Create** — the SSH key is created. *(SAP recommends a 4096-bit key size.)*
3. Click **Actions → Download Public (OpenSSH) Key** — you get an `<alias>.pub` file. **Done.**
4. Send that **public key** to the SFTP server team; they add it as an authorised key for your user.
5. In the SFTP adapter choose **Authentication = Public Key** and give the **Private Key Alias** (the alias from step 2) and the user name.

5. Testing the Connection and the Known Hosts File

To check the connection: **Monitor tab → Connectivity Test → SSH** → enter the host (e.g., `tcs265.files.com`) → provide all required fields → **test the connection**. In class the test was run with **Public Key, User Credentials and None**.

Why the "Known Hosts" File?

Generally the SFTP side will ask **what is required for the SFTP connection** — the answer is the **known_hosts file**.

> **What is a known-hosts file?** As per the name: for whichever SFTP server you are connecting to, it tells you whether that server is **known (a valid SFTP server) or not**. It is how CPI **identifies the SFTP server**.
>
> **Earlier case (PI/PO):** we used the **server fingerprint**. **Now in CPI:** instead of the server fingerprint, we use the **host key** — "getting the host key fingerprint".

Steps to Get the Host Key and Deploy It

1. **Monitor tab → Connectivity Test → SSH** → provide all required details → **Authentication = None** → click **Send**.
2. **Result:** Host Key Type = **ssh-rsa**; the **Host Key / fingerprint is generated → copy it**.
3. Go to **Security Material → Create → Known Hosts → Add → paste** the host key → **Deploy**.

> **Two refinements (verified in SAP's documentation):**
> - Use the **"Copy Host Key"** button under *Response* — that copies the full host-key entry to paste into the known-hosts file. The fingerprint is the short code you **compare with the SFTP administrator** to be sure you are trusting the real server (don't blindly trust the first key you receive).
> - The artifact name is **known_hosts** (with an underscore).

Host Key Verification — Three Options

Option	Meaning
Against Tenant	The server's host key is checked against the **known_hosts artifact deployed on your tenant** — **iFlows use this by default** (per class)
Against Partner Directory	The host key is checked against known hosts stored in the Partner Directory (useful when many partners each have their own server)
Off	No verification — insecure, use only for quick testing

Check Directory ☑ — `/output`**:** tick **Check Directory Access** to confirm the user is allowed to access that folder (if left empty, the default is the user's home directory).

> **Main important point (flagged "mandatory" in your notes): without placing the host key in the known_hosts artifact, we cannot use the SFTP adapter in iFlows** (with host-key verification on). Anyone can connect to a public SFTP server — but **from CPI you need the known_hosts entry (host key fingerprint)**; it helps to **identify whether the host is valid or not.**

Part B — Gather with Zip, and Zip Compression

6. Recap — Gather's Aggregation Strategy (Day 24)

Gather → Aggregation Strategy: Incoming Format (Any · Plain Text · XML Different Format · XML Same Format) and **Aggregation Algorithm**. Today's focus: the **Zip** algorithm.

7. Gather with the Zip Algorithm

> **Zip algorithm:** if you want to **compress files and place them as a zip file**, you use this option.

Demo

<table>
<tr><td rowspan="2">Postman → HTTPS</td><td rowspan="2">Parallel Multicast</td><td>CM1 (Emp ①)</td><td rowspan="2">→ Join → Gather (Zip) → End</td><td rowspan="2">SFTP → Receiver</td></tr>
<tr><td>CM2 (Emp ②)</td></tr>
</table>

Setting	Value in Demo
Gather → Incoming Format	XML (Same Format)
Gather → Aggregation Algorithm	**Zip**
Gather → File Name	`test.xml` — the name given to the entry inside the zip
SFTP adapter → File Name	`Employees.zip` — the name of the zip file itself

What Gather with Zip does: 1 zip file → multiple files placed inside it.

> **Duplicate-name trap (verified in SAP's documentation):** the Gather's **File Name** field takes an expression such as `${header.id}.xml`, and **the name must be unique within the archive** — a zip cannot hold two files with the same name, so processing raises an error. If every message used the fixed name `test.xml`, the two employees would clash. Set a per-message header **before** the Gather (e.g., the employee ID) and use it in the file name — which also matches your later output example `Employee_<id>.xml`. If the field is empty, CPI uses the `CamelFileName` header, or generates a unique name.

8. Zip Compression — Separate Zip per Record

New requirement: separate XML records → separate XML file → separate zip file. What now? Use the palette function **Zip Compression**.

<table>
<tr><td rowspan="2">Sender → HTTPS → Start</td><td rowspan="2">Parallel Multicast</td><td>Branch 1: CM (Emp 1 details) → Zip Compression → End</td><td rowspan="2">SFTP → Receiver</td></tr>
<tr><td>Branch 2: CM (Emp 2 details) → Zip Compression → End</td></tr>
</table>

> **Zip Compression:** if you want to **compress a message and place it as a zip file**, use Zip Compression. **Zip Compression → no configuration details** — you just drop it in.
>
> **Two points from class:**
>
> - **The SFTP server has to be called two times** (once per branch/file).
> - **We should not use Join when we want to use Zip Compression** — Join would merge the branches into one path, but here each branch must finish on its own.

> **Verified:** SAP's blog confirms the ZIP Encoder (palette: *Message Transformers → Encoders → ZIP Compression*) has **no configuration parameters** — it simply zips a **single message** into a zip archive.

Which One Do I Use?

Situation	Use
Several messages/files → **one** zip archive	**Gather → Zip** format

One message → zip that single file	**Zip Compression** (Encoder)

9. Reading a Zip File — Zip Decompression

What if I want to read a zip file? → decompress it. **Palette → Transformation → Decoder → Zip Decompression → unzips the zip file.**

Scenario: SFTP sender → file picked (a zip file) → Zip Decompression → **output:** `Employee_<id>.xml` (e.g., one employee XML file).

> **Important limit (verified in SAP's blog):** the **ZIP Decoder handles only a zip with a single entry — if the archive contains more than one file, an error is raised.** For a **multi-file** zip use the **Zip Splitter** (palette: *Routing → Splitter → Zip Splitter*, Day 23). It splits the archive into one message per file and has two settings: **File Name Pattern** (e.g., `*.xml` to process only XML files) and **Stop on Exception**.

Task	Step
Zip **one** message	Encoder → Zip Compression
Zip **many** messages into one archive	Gather → Zip
Unzip an archive with **one** file	Decoder → Zip Decompression
Unzip an archive with **many** files	Splitter → Zip Splitter

10. Quick Interview Answers

Question	Short Answer
Why SFTP in CPI and not the File adapter?	The File adapter reads local on-premise folders (PI/PO); CPI is cloud-based and has no local folders, so it reaches files via SFTP
What is public-key authentication?	You keep the private key; the SFTP server holds your public key and uses it to verify your requests
What is Dual authentication?	Public key plus user name/password together
What is the known_hosts file?	The list of trusted SFTP server host keys deployed on the tenant; CPI checks the server against it
How do you get the host key?	Monitor → Connectivity Test → SSH (authentication None) → Copy Host Key → add to Known Hosts → Deploy
Gather–Zip vs. Zip Compression?	Gather–Zip: many messages into one archive. Zip Compression: zips a single message
Should I use Join with Zip Compression?	No — each branch finishes separately (and calls SFTP separately)
How do I read a zip file?	Zip Decompression (single file) or Zip Splitter (many files)

> **Two notes I couldn't fully interpret:**
>
> - On the last page the output line starts with "**PC →**" before `Employee_id.xml`. I read this as "the output is one file per employee, e.g. `Employee_<id>.xml` " — tell me if your trainer meant something else.
> - Page 12's clarification question ("where do you use Gather → zip format and Zip compression") is answered by the table in section 8 — I answered it from the demos rather than guess any extra point.